



Magic Tree: BST Notes



The Memory Story (Library Connection)

Socho ek badi si library hai jahan bohot saari books hain:

- **Normal Binary Tree:** Ek messy library hai jahan max 2 books ek shelf par hain, par koi order nahi. Book dhoondhne mein bohot time lagta hai $O(n)$.
- **BST (Binary Search Tree):** Ek smart, magical library! Yahan ek **Golden Rule** chalta hai jisse searching ekdam fast ho jati hai $O(\log n)$.



THE GOLDEN RULE



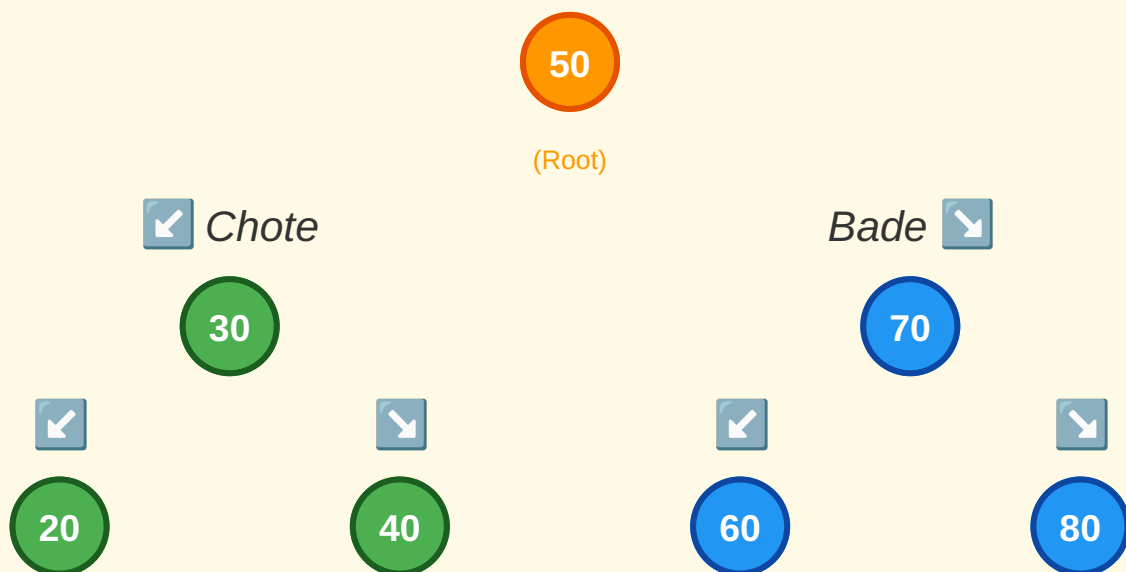
Left wale hamesha CHOTE (Small) 🧑

Right wale hamesha BADE (Big) 🧑



Homemade Colorful Diagram

Let's insert 50, 30, 70, 20, 40, 60, 80





Hide & Seek (Insert & Search)

- **Search (Dhoondhna):** Root se pucho. Agar value choti hai toh Left jao, badi hai toh Right jao. Agar chalte-chalte raasta khatam (None) ho gaya, matlab element nahi hai!
- **Insert (Naya element daalna):** Same Search wala raasta follow karo. Jahan end mein khali jagah (None) mile, wahan naya node (block) chipka do.
- **Speed:** **$O(\log n)$** kyunki har step par hum aadhi branch ko ignore kar dete hain.



The Magic Spell (Interview Secret Trick)

Interviewer aapko confuse karne ke liye 3 sawaal poochega:

1. BST ko increasing (sorted) order mein kaise print karein?
2. Kaise check karein ki tree valid BST hai ya nahi?
3. BST ka In-order traversal kya hota hai?

✨ **TEENO KA ANSWER EK HI HAI** ✨

IN-ORDER TRAVERSAL (Left → Root → Right)

Memory Link: Jab bhi hum In-order traversal karte hain, BST automatically elements ko **Sorted (chote se bada)** order mein nikal kar deta hai! Output check karo, agar sorted hai toh tree valid hai.